

Functional Annotation of PP_1969 (MoaA) in *Pseudomonas putida* KT2440

Gene: PP_1969 (OrderedLocusName) · **UniProt:** Q88LG4 · **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / **KT2440**), NCBI TaxID PSEPK

1. Summary (answer to the research question)

PP_1969 encodes **MoaA**, a cytoplasmic **radical S-adenosyl-L-methionine (radical SAM) metalloenzyme** that catalyzes the **first committed step of molybdenum cofactor (Moco) biosynthesis**. Its primary function is that of a **GTP 3',8-cyclase (EC 4.1.99.22)**: it converts **guanosine-5'-triphosphate (5'-GTP)** into the cyclic intermediate **3',8-cyclo-7,8-dihydro-GTP (3',8-cH₂GTP)**, which the partner enzyme MoaC then converts into **cyclic pyranopterin monophosphate (cPMP, "precursor Z")**. The enzyme carries **two [4Fe-4S] clusters** — an N-terminal radical-SAM cluster that reductively cleaves SAM to generate a 5'-deoxyadenosyl radical, and a MoaA-unique C-terminal ("twitch") cluster that binds and activates the GTP substrate. MoaA acts in the **cytoplasm**, feeding the pterin scaffold that is ultimately matured into the molybdenum cofactor required by all of the cell's molybdoenzymes (e.g., nitrate reductase, sulfite oxidase, xanthine/aldehyde oxidoreductases, DMSO/TMAO reductases).

Identity is unambiguous: the protein's domain set (radical SAM, MoaA-like, MoaA_twitch, TIM-barrel), its sequence motifs, and its KEGG orthology (K03639) all define a canonical MoaA. **The gene symbol is NOT ambiguous** for this entry.

2. Gene/Protein identity verification

Criterion	Finding
Symbol vs. description	"Molybdenum cofactor biosynthesis protein A" $\Rightarrow$ <i>moaA</i> ; consistent.
Organism	<i>P. putida</i> KT2440 (PSEPK); confirmed via UniProt/KEGG (ppu:PP_1969).
Domains/family	Radical SAM core (IPR006638), MoaA-like (IPR040064), MoaA_twitch (IPR010505), MoCo_biosynth_MoaA/MoaC (IPR050105), TIM-barrel (IPR013785) — the exact diagnostic set of the MoaA family.
Literature match	Extensive primary literature on MoaA (bacterial) / MOCS1A (human ortholog) aligns precisely with the domains and reaction.

Sequence-level confirmation (Q88LG4, 337 aa, 36.7 kDa; this work, UniProt): - N-terminal radical-SAM **CX₃CX₂C** motif: **C36-N-Y-A-C40-T-Y-C43**. - C-terminal "twitch"-domain cluster cysteines: **Cys268, Cys271, Cys285** (C-R-T-C...W-L-H-G-C) — a three-Cys ligation leaving a unique, substrate-binding Fe. - Conserved C-terminal **di-glycine (GG) motif**: the protein ends in **...KVIGG**. - UniProt keywords: 4Fe-4S, GTP-binding, Iron-sulfur, S-adenosyl-L-methionine, Molybdenum cofactor biosynthesis, Lyase. **No signal peptide or transmembrane segment.**

All of these are hallmark, functionally essential features of MoaA, so the identification is secure both by homology/domain analysis and by direct motif inspection.

3. Primary function: the reaction catalyzed and substrate specificity

Reaction (step 1 of Moco biosynthesis):

$5'\text{-GTP} + \text{SAM} + 2e^- \rightarrow 3',8\text{-cyclo-7,8-dihydro-GTP (3',8-cH}_2\text{GTP)} + 5'\text{-deoxyadenosine} + \text{L-methionine}$ (MoaC subsequently: $3',8\text{-cH}_2\text{GTP} \rightarrow \text{cyclic pyranopterin monophosphate, cPMP / precursor Z}$)

- **The substrate is intact 5'-GTP**, bound with high affinity; nucleotide-specificity determinants were mapped biochemically and structurally [Hänzelmann & Schindelin 2006,

[P 16632608](#)];

2004,

[P 15317939](#)].

- The reaction is a **radical-initiated intramolecular rearrangement** that inserts the **guanine C8 carbon** into the new pyranopterin ring — a chemically demanding C–C bond formation characteristic of radical SAM chemistry [Pang & Yokoyama 2018,

[P 30097104](#)];

Yokoyama 2017,

[P 29072833](#)].

- The classical description of "MoaA + MoaC convert GTP → precursor Z" was refined by Yokoyama and colleagues, who showed **MoaA alone acts as a GTP 3',8-cyclase producing 3',8-cH₂GTP**, with MoaC completing cPMP formation [Hover et al. 2015,

[P 25697423](#)];

review Yokoyama, Li & Pang 2022,

[P 35480226](#)].

EC number: 4.1.99.22 (GTP 3',8-cyclase). UniProt classifies molecular function as **Lyase**, consistent with C–C bond formation/cyclization.

4. Cofactors and catalytic mechanism (structural evidence)

MoaA is a **homodimer**; each subunit is built on an **incomplete (β/α) TIM-barrel** and carries **two [4Fe-4S] clusters ~17 Å apart** flanking a large active-site pocket [Hänzelmann & Schindelin 2004,

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P 15317939

1. **N-terminal radical-SAM cluster** ($CX_3CX_2C = \text{Cys36/40/43}$): SAM chelates the unique Fe as an N/O chelate and is **reductively cleaved to give L-methionine and a 5'-deoxyadenosyl radical (5'-dA•)** [PMID 15317939;

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P 16632608

2. **C-terminal MoaA-unique "twitch" cluster** (Cys268/271/285): its **non-cysteinyl-ligated unique Fe binds the substrate**. The crystal structure with 5'-GTP and ENDOR spectroscopy showed the **guanine N1 nitrogen coordinates this Fe**, positioning the substrate and preventing escape of radical intermediates via a tightly anchored triphosphate [Hänzelmann & Schindelin 2006,

;

P 16632608

Lees et al. 2009,

].

P 19566093

3. The **conserved C-terminal GG motif** (...KVIGG in PP_1969) folds into the SAM-binding pocket and is **essential for SAM binding and radical initiation**; its mutation abolishes activity and, in the human ortholog MOCS1A, causes Moco deficiency [Hover et al. 2015,

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P 25697423

The mechanism and residues are strongly conserved between bacterial MoaA and the human ortholog **MOCS1A** — indeed **>50 % of human Moco-deficiency (MoCD) patients carry MOCS1A mutations**, underscoring the reaction's conservation and importance [Hover et al. 2015,

;

P 25697423

Hänzelmann & Schindelin 2004,

].

P 15317939

5. Localization

MoaA carries out its function in the **bacterial cytoplasm**: - It has **no signal peptide and no transmembrane segments** (Q88LG4 keywords/features) and the prototype is a soluble homodimer purified/crystallized from cell lysate [PMID 15317939]. - Its two **oxygen-labile [4Fe-4S] clusters** and the **oxygen-sensitive tetrahydropyranopterine product** require the **reducing cytoplasmic environment**; the Moco biosynthesis network relies on **protected metabolic channelling** of these oxygen-sensitive intermediates between enzymes [Hänzelmann & Schindelin 2006,

P 16632608

Kaufholdt et al. 2017,

P 29184564

6. Pathway context in *P. putida* KT2440

Moco biosynthesis is a **conserved four-step bacterial pathway** [Mendel & Leimkühler 2015,

P 24980677

1. **GTP → cPMP — MoaA (PP_1969)** [step 1a, this enzyme] → **MoaC** [step 1b];
2. **cPMP → molybdopterin (MPT)** — MPT synthase **MoaD/MoaE** (+ activating enzyme MoeB);
3. **Mo insertion into MPT → Moco — MogA/MoeA**;
4. **(bacteria) nucleotide addition to form the dinucleotide (MGD) variant — MobA**.

In *P. putida* KT2440, KEGG mapping (this work) places the downstream partners in a **separate genomic cluster** — **moaC = PP_1292, moaD = PP_1293, moaB = PP_1294**, with **moeA = PP_2123** and **mogA = PP_3457** — whereas **PP_1969 is not in a classical *moaABCDE* operon** (its neighbors are an unrelated DNA-pol-III subunit PP_1966, tRNA D-aminoacylase PP_1967, a TetR-family regulator PP_1968, and a lipoprotein PP_1970). The GTP-3',8-cyclase orthology group (KO **K03639**) additionally contains **three *P. putida* loci** — **PP_1969, PP_2482, PP_4597** — indicating MoaA-family/radical-SAM paralogs; PP_1969 is the reference-annotated *moaA*.

Physiological purpose: the Moco produced downstream is the catalytic cofactor of the cell's **molybdoenzymes** (nitrate reductases, sulfite oxidase, xanthine/aldehyde oxidoreductases, DMSO/TMAO reductases, etc.), which are central to redox and anaerobic/assimilatory

metabolism [Mendel & Leimkühler 2015, P 24980677];
 Schwarz, Mendel & Ribbe 2009, P 19675644].
 MoaA thus sits at the **entry point** of this essential cofactor supply line.

7. Supported and refuted hypotheses

Supported: - H1 — *PP_1969 is a MoaA ortholog (GTP 3',8-cyclase), not a metabolic aldolase despite the TIM-barrel annotation.* Supported by domain set, KEGG K03639, and all catalytic motifs. - H2 — *It is a radical-SAM enzyme with two [4Fe-4S] clusters.* Supported by CX₃CX₂C + three-Cys twitch motifs and UniProt Fe-S/SAM keywords, mirroring the crystallographically characterized prototype. - H3 — *Its substrate is 5'-GTP and its product is 3',8-CH₂GTP (→ cPMP with MoaC).* Supported by biochemical/structural literature. - H4 — *It functions in the cytoplasm.* Supported by absence of signal/TM features and O₂-sensitivity of clusters and product.

Refuted / set aside: - The literal "Aldolase_TIM" (IPR013785) annotation as a *functional* aldolase is misleading — the TIM-barrel is a structural scaffold, not evidence of aldolase activity. - The older idea that MoaA+MoaC jointly synthesize precursor Z in one indivisible step is superseded: MoaA alone is a GTP 3',8-cyclase (product 3',8-CH₂GTP).

8. Evidence quality and limitations

- **Direct experimental evidence** (crystallography, ENDOR/EPR spectroscopy, in-vitro reconstitution, mutagenesis) exists for the **MoaA family prototype** (from *Staphylococcus aureus*/*E. coli* systems) and the human ortholog MOCS1A — not specifically for the *P. putida* PP_1969 protein. Given the very high conservation of sequence, domains, and the essential motifs verified here, transfer of function to PP_1969 is well justified (inference from homology/structure), but a *P. putida*-specific biochemical study was not identified.

- The three K03639 paralogs (PP_1969/2482/4597) raise the possibility of functional redundancy or specialization; distinguishing their individual roles would require targeted knockouts/complementation, not available here.
- No experimental subcellular-localization study specific to PP_1969 was found; cytoplasmic assignment is a well-supported inference.

9. Future directions

- In-vitro reconstitution of PP_1969 to confirm GTP 3',8-cyclase activity and kinetics.
 - Genetic dissection of the three *P. putida* K03639 paralogs (single/multiple deletions; Moco-dependent phenotypes such as nitrate reductase activity).
 - Test whether PP_1969 physically channels its product to MoaC (PP_1292) despite genomic separation.
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Key references

- Hänzelmann P, Schindelin H. *PNAS* 2004 (PMID **15317939**) — MoaA crystal structure; identification of 5'-GTP as substrate; two [4Fe-4S] clusters; MoCD relevance.
 - Hänzelmann P, Schindelin H. *PNAS* 2006 (PMID **16632608**) — MoaA·5'-GTP structure; N-terminal cluster cleaves SAM → 5'-dA•; C-terminal cluster binds substrate.
 - Lees NS, Hänzelmann P, ... Hoffman BM. 2009 (PMID **19566093**) — ENDOR: guanine N1 binds cluster II.
 - Hover BM, ... Yokoyama K. *JACS* 2015 (PMID **25697423**) — MoaA = GTP 3',8-cyclase producing 3',8-cH₂GTP; essential C-terminal GG motif; MOCS1A/MoCD link.
 - Pang H, Yokoyama K. 2018 (PMID **30097104**); Yokoyama K, Li & Pang 2022 (PMID **35480226**); Yokoyama K 2017 (PMID **29072833**) — mechanism reviews.
 - Mendel RR, Leimkühler S. 2015 (PMID **24980677**); Schwarz, Mendel & Ribbe 2009 (PMID **19675644**); Kaufholdt et al. 2017 (PMID **29184564**) — Moco pathway/network reviews.
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